

ADAM — Custom PCB Reference Document

Pi Zero 2W-Only Architecture | Body + Head Motherboards | Two Microphone BOM Variants

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Document revision: v1.0 | Consolidated from BOM1 (Pi Zero 2W, rev V1.2) + Head Board (ESP32-S3, rev V1.1, strap-pin corrected) | BOM2 (CM4) intentionally dropped per direction: Pi Zero 2W only.

Purpose: single authoritative reference for schematic capture — every net, connector, IC pin, and BOM line consolidated in one place, in two buildable microphone configurations.

0. Verdict on the Updated BOM1 (Pi Zero 2W Body Board, rev V1.2)

Short answer: **the V1.2 update is a real, substantive hardening pass** — it directly fixes three of the issues flagged in the prior review. It is not yet a complete package because two items from that review were never revisited (they live on the Head board and the now-dropped CM4 board), and one new documentation gap appears in the V1.2 text itself. Detail below.

Confirmed fixed in V1.2 (Body / Pi Zero 2W board)

- **Servo point-of-load protection.** V1.2 adds a dedicated 1.5 A PTC (F_SERVO), a local 1000µF low-ESR capacitor at the servo header, and an SS34 Schottky flyback diode across +5V_SERVO/GND, all placed at J_SERVO itself rather than relying on the shared 2200µF input reservoir. This directly answers the prior finding that a single upstream cap does not protect against trace-resistance sag at the point of load — and it matters more than usual here because servo-induced brownout reboots are a *documented, already-occurred failure* in this project's history (ENABLE_SERVOS=0 was shipped as a workaround).
- **MAX98357A GAIN pin.** Now explicitly tied to GND via a 100kΩ resistor for a fixed +12dB, with the connection spelled out in both the BOM line and the netlist table. This is the exact defect (floating GAIN → drifting volume) already observed and documented in the current build's hardware-architecture notes, so nailing it down in the new PCB is the correct and necessary fix.
- **PCM1808 clock topology and mode strapping.** V1.2 explicitly declares the Pi as I2S bus master, drives PCM1808 SCKI from Pi GPCLK0 (GPIO4) at 12.288 MHz (256×48kHz) instead of leaving the ADC free-running off its own crystal, and adds explicit FMT/MD0/MD1/BYPAS strapping resistors. This resolves the dual-clock-domain risk (audio clicks/dropouts from an unsynchronized ADC clock) flagged previously, and removes the ambiguity around PCM1808 mode configuration.
- **Neck cable spec.** Now explicitly a shielded, twisted 28AWG high-flex cable with single-point shield grounding at the body side — addresses the earlier note that cable spec was left implicit despite this project's known sensitivity to mechanical/electrical noise coupling on the neck harness.

Still open — not addressed by this update

- **Head board (ESP32-S3) strapping-pin conflict — UNCHANGED, still present.** The uploaded Head board file is textually identical to the version reviewed previously (same rev V1.1). GPIO45 (VDD_SPI strap) still drives TFT_DC, GPIO46 (boot-log strap) still drives TFT_CS, and GPIO3 (JTAG-source strap) still drives TOUCH3 — the exact contradiction of the document's own 'zero strapping conflicts' claim. This document reassigns these nets (Section 5) so the deliverable below is corrected, but note for the team: the source .md file itself has not been updated to reflect this.
- **CM4 board issues are moot, not resolved** — dropped per this session's direction to build Pi Zero 2W only. If a CM4 variant is revisited later (e.g. for on-device vision), the missing U.FL/antenna footprint and missing reflash/debug header from the prior review still apply and were never fixed.
- **Mic architecture decision — now correctly externalized rather than buried.** V1.2 keeps the discrete electret + PCM1808 signal chain as its only BOM. Per this session's direction, that decision is no longer silently forced: this document ships **two full BOM variants** (Section 3: INMP441 digital module → preserves every line of the already-tuned software audio pipeline; Section 4: discrete electret + PCM1808 → matches the V1.2 .md exactly, with its new clock/strapping fixes carried through). Recommendation stands: default to the INMP441 variant unless there is a specific cost/supply reason to run the analog chain, because the entire `audio\_utils.py` gate/floor/shape-vote tuning history in this project assumes I2S digital mics with known bit alignment.

One new item introduced by V1.2 itself

The servo branch adds F_SERVO (1.5 A PTC) *in series* with the servo supply, downstream of the 1000µF local reservoir. Series resistance/impedance from a PTC in its hold state (typically 0.1–0.3Ω for a part this size) sits between the reservoir cap and the load, which mildly undercuts the cap's ability to answer a fast current transient right at the connector. This is a minor point, not a blocker — the correct fix, carried into Section 2 below, is to place **C_SERVO_LOC after F_SERVO, directly at J_SERVO's power pins**, so the fuse never sits between the reservoir and the motor. Verify PTC placement against this note during layout.

1. System Architecture Overview (Pi Zero 2W Only)

Two boards, joined by a shielded 4-wire neck harness (+5V, GND, TX, RX @ 921,600 baud). The BODY board carries the Raspberry Pi Zero 2W and everything audio/servo/power. The HEAD board carries the ESP32-S3 (camera, TFT display, touch, tilt servo) as a single consolidated MCU, replacing the legacy ESP32-CAM + Pico two-chip head.

Subsystem	Description
BODY — Compute	Raspberry Pi Zero 2W (BCM2710A1, Quad Cortex-A53 @ 1GHz, 512MB RAM, Wi-Fi/BT)
BODY — Audio In (Variant A)	2× INMP441 I2S digital MEMS mic modules → direct I2S to Pi (no analog front end)
BODY — Audio In (Variant B)	2× electret capsules → LMV358 dual op-amp preamp → PCM1808 24-bit/96kHz I2S ADC (Pi is I2S bus master; PCM1808 SCKI synced from Pi GPCLK0)
BODY — Audio Out	MAX98357A Class-D I2S amp (GAIN fixed +12dB via 100kΩ to GND) → 3W 4/8Ω speaker
BODY — Actuation	MG90S pan servo on dedicated point-of-load protected branch (local 1000μF + flyback diode + isolated PTC)
BODY — Presence Sensing	RCWL-0516 microwave radar → digital GPIO input
BODY — Head Link	4-pin shielded JST-PH (+5V, GND, UART TX, UART RX) @ 921,600 baud, single-point shield ground at body
HEAD — Compute	ESP32-S3-WROOM-1-N8R8 (dual Xtensa LX7 @ 240MHz, 8MB Octal PSRAM) — camera + display + touch + tilt, unified
HEAD — Vision	OV2640 2MP camera, 24-pin 0.5mm FPC, DVP parallel bus
HEAD — Display	2.4" 320×240 ILI9341 TFT, 20-pin 0.5mm FPC, SPI (DMA)
HEAD — Touch	4× TTP223 capacitive modules (cheeks ×2, petting ×2) via JST-PH, or native ESP32-S3 touch pads (populate one or the other, not both)
HEAD — Actuation	MG90S tilt servo, local 220μF reservoir at servo header
HEAD — Programming	8× pogo-pin pads on PCB underside: 3V3, GND, EN, IO0, TX0, RX0, USB D-, USB D+

2. Body Motherboard (Pi Zero 2W) — Subsystems Common to Both Mic Variants

These sections are identical regardless of which microphone BOM (Section 3 or 4) is populated. Designators reused across variants keep the same reference names for schematic sheet consistency.

2.1 Compute & Mounting

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
U1	Raspberry Pi Zero 2W	SC0510 / RPI-ZERO-2W	Module, 65×30mm	1	Rs. 2,150.00	Robu.in / Supreme Electronics (Chandni Chowk)
J1A, J1B	2×20 Dual-Row Female SMD Header, 2.54mm pitch, 5.0mm low profile	2X20-FEMALE-SMD-2.54	SMD Dual Row 2×20	1	Rs. 35.00	Quartz Components / Robu.in

2.2 Power Input & Protection

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
J_PWR	USB Type-C Receptacle, power-only, 5V@3-5A	TYPE-C-16P-SMD	SMD Hybrid 16-Pin	1	Rs. 18.00	Evelta / Robu / Chandni
R1, R2	5.1kΩ 1% resistor — USB-C CC1/CC2 pulldown	RC0603FR-075K1L	0603	2	Rs. 1.50	Local passives / Robu
F1	PTC resettable polyfuse, 3.0A hold / 3.5-6.0A trip, 6V min — primary input	SMD1812P300TF	1812	1	Rs. 16.00	Evelta / Mouser
Q1	P-Channel MOSFET, 30V 4.2A, Rds(on)<50mΩ — reverse-polarity protection	AO3401A / DMG3415U	SOT-23-3	1	Rs. 6.00	Quartz / Evelta
D1	TVS diode array, ESD protection USB/VBUS, 5V unidirectional	USBLC6-2SC6	SOT-23-6	1	Rs. 12.00	Evelta / Mouser
C_BULK_IN	2200μF 16V low-ESR radial electrolytic — input power reservoir	UPW1C222MHD	Radial 10×20mm	1	Rs. 24.00	Supreme / Classic Electronics
C1,C2,C3	10μF 16V X5R ceramic decoupling	CL10A106KP8NNNC	0603	3	Rs. 3.50	Local passives / Robu
C4,C5,C6	100nF 50V X7R ceramic decoupling	CC0603KRX7R9BB104	0603	3	Rs. 1.50	Local passives / Robu

2.3 Servo Point-of-Load Protection (Pan Servo Branch)

Placement order from +5V_SYS to J_SERVO: **F_SERVO** (isolation PTC) → **C_SERVO_LOC** (reservoir cap, placed at connector pins, after the fuse) → **J_SERVO**, with D_SERVO (flyback diode) across the connector pins in parallel with C_SERVO_LOC. This ordering (cap after fuse, right at the pins) is the correction noted in Section 0 vs. the as-written V1.2 sequence.

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
F_SERVO	PTC resettable polyfuse, 0.75A hold / 1.5A trip — servo subsystem isolation	SMD1206P075TF	1206	1	Rs. 9.00	Evelta / Mouser
C_SERVO_LOC	1000μF 16V low-ESR radial electrolytic, mounted at J_SERVO pins	UPW1C102MPD	Radial 8×15mm	1	Rs. 14.00	Supreme Electronics

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
D_SERVO	3A 40V Schottky diode — flyback/back-EMF clamp, cathode to +5V_SERVO, anode to GND	SS34 / B340A	SMA / DO-214AC	1	Rs. 5.00	Local passives / Robu
J_SERVO	3-pin JST-XH/Berg header — GND, +5V_SERVO, PWM (to MG90S pan servo)	B3B-XH-A(LF)(SN)	2.54mm R/A	1	Rs. 8.00	Robu / Chandni

2.4 Audio Output (Speaker Amplifier — Common to Both Variants)

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
U_AMP	3.2W mono Class-D I2S audio amp with internal DAC	MAX98357AETE+T	16-TQFN 3×3mm	1	Rs. 125.00	Evelta / Mouser / DigiKey
R_GAIN	100kΩ 1% — GAIN_SLOT (Pin 2) tied to GND → fixed +12dB (floating-pin drift fix)	RC0603FR-07100KL	0603	1	Rs. 1.50	Local passives
L_FLT1,L_FLT2	Ferrite bead, 100MHz 600Ω, 2A — speaker EMI filter	BLM18PG601SN1D	0603	2	Rs. 6.00	Mouser / Evelta
C_FLT1,C_FLT2	220pF 50V C0G ceramic filter cap	CC0603JRNPO9BN221	0603	2	Rs. 2.00	Local passives
C_AMP	10μF 16V X5R ceramic — MAX98357A VDD bulk	CL10A106KP8NNNC	0603	1	Rs. 3.50	Local passives
J_SPK	2-pin JST-PH — to 3W 4/8Ω speaker	S2B-PH-K-S(LF)(SN)	2.0mm R/A	1	Rs. 7.00	Robu / Quartz

2.5 Peripherals & Head-Link Connector

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
J_RADAR	3-pin JST-PH — GND, +5V, OUT (to RCWL-0516 microwave radar)	S3B-PH-K-S(LF)(SN)	2.0mm R/A	1	Rs. 7.00	Robu / Quartz
J_NECK	4-pin JST-PH shrouded — 5V, GND, TX, RX (to ESP32-S3 Head board)	S4B-PH-K-S(LF)(SN)	2.0mm R/A	1	Rs. 9.00	Robu / Quartz
CBL_NECK	4-core shielded high-flex twisted cable, 28AWG silicone, braided shield to GND (single-point, body side only)	CBL-4C-SHIELD-28AWG	External harness, 30cm	1	Rs. 35.00	Evelta / Chandni wire market
LED1	Green LED — 5V system power indicator	0603-LED-GREEN	0603	1	Rs. 2.00	Local passives
R_LED1	1.0kΩ 1% — LED current limiter	RC0603FR-071KL	0603	1	Rs. 1.50	Local passives
TP1-TP8	SMD test points: GND, 5V, 3V3, BCLK, WS, DIN, DOUT, PWM	RC0805 pad	0805 copper pad	8	Rs. 0.00	PCB etched copper

3. Microphone BOM — Variant A: INMP441 I2S Digital Module (Recommended)

Recommended default. Two ready-made INMP441 breakout modules, each a self-contained I2S digital MEMS microphone (24-bit, 48kHz native). No analog preamp, no external ADC, no clock-sync design problem — the modules are I2S slaves driven directly by the Pi's I2S bus, exactly matching the microphone architecture the existing, extensively field-tuned audio pipeline (gate thresholds, per-channel noise-floor learning, S32-shift gain staging, DOA via GCC-PHAT) was built and debugged against. Selecting this variant means zero software audio-chain rework is required when the custom PCB replaces the current breadboard/HAT build.

Stereo wiring convention

Both INMP441 modules share BCLK and WS (I2S clock lines are bused). Each module's L/R select pin sets which half of the I2S frame it occupies: tie one module's L/R to GND (occupies the Left slot) and the other's L/R to 3.3V (occupies the Right slot). Both modules' SD (data out) pins tie to the same Pi I2S_DIN net — the INMP441 tri-states its output during the half-frame it doesn't own, so this is a safe bus, not a conflict.

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
J_MIC_L	4-pin JST-PH — VDD, GND, SD, WS/LRSEL (to Left INMP441 breakout)	S4B-PH-K-S(LF)(SN)	2.0mm R/A	1	Rs. 9.00	Robu / Quartz
J_MIC_R	4-pin JST-PH — VDD, GND, SD, WS/LRSEL (to Right INMP441 breakout)	S4B-PH-K-S(LF)(SN)	2.0mm R/A	1	Rs. 9.00	Robu / Quartz
MIC_L, MIC_R	INMP441 I2S digital MEMS microphone breakout module (24-bit, omnidirectional, on-board regulator)	INMP441-BREAKOUT	6-pin breakout, external	2	Rs. 180.00 (Rs. 90 ea)	Robu.in / Evelta
R_LRL	10kΩ 1% — Left module L/R pulldown to GND (selects Left I2S slot)	RC0603FR-0710KL	0603	1	Rs. 1.50	Local passives
R_LRR	10kΩ 1% — Right module L/R pullup to 3V3 (selects Right I2S slot)	RC0603FR-0710KL	0603	1	Rs. 1.50	Local passives
C_MICVDD1, C_MICVDD2	100nF 50V X7R — local VDD decoupling at each mic connector	CC0603KRX7R9BB104	0603	2	Rs. 1.50	Local passives

Variant A — Pin Map (Pi Zero 2W header)

Pi Header Pin	BCM GPIO	Net Name	Connected To	Function
Pin 1	—	+3V3_PI	MIC_L VDD, MIC_R VDD	3.3V mic supply (INMP441 has on-board LDO tolerant of 1.8-3.3V; use 3V3 for simplicity)
Pin 6/9/14/20	—	GND	MIC_L GND, MIC_R GND	Common ground return
Pin 12	GPIO18	I2S_BCLK	MIC_L BCLK, MIC_R BCLK (bused)	Shared I2S bit clock, Pi as I2S master
Pin 35	GPIO19	I2S_WS	MIC_L WS, MIC_R WS (bused)	Shared word-select / frame clock
Pin 38	GPIO20	I2S_DIN	MIC_L SD, MIC_R SD (bused, tri-state per slot)	Digital audio input from both mics, stereo interleaved

Variant A — Cost Summary (Microphone Front End Only)

Subsystem	Included	Est. Cost (INR)
Mic Input Subsystem (Variant A)	2× INMP441 modules, JST connectors, LR-select pulldowns, decoupling	Rs. 202.50

4. Microphone BOM — Variant B: Discrete Electret Capsules + On-Board PCM1808 ADC

Matches the updated BOM1 (.md, rev V1.2) exactly, including its V1.2 clock-sync and strapping fixes. Choose this variant only if there is a specific cost, supply-chain, or BOM-simplification reason to avoid a ready-made I2S mic module (e.g. INMP441 breakout availability/pricing at volume) — because selecting it means the existing tuned audio-gate software (noise-floor learning, spectral-shape gate, DOA) will need to be re-validated against a new analog signal chain with different self-noise, DC bias, and gain-staging behavior.

4.1 Analog Front End (Preamp)

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
J_MIC_L, J_MIC_R	2× 2-pin JST-PH — Left/Right electret capsule input	S2B-PH-K-S(LF)(SN)	2.0mm R/A	2	Rs. 7.00	Robu / Quartz
MIC_CAP	Electret condenser mic capsule, 6/9mm omnidirectional	C9767 / KECG2742	2-pin lead, external	2	Rs. 9.00 ea	Supreme Electronics / Chandni
U_PREAMP	Low-noise dual rail-to-rail op-amp — mic preamplifier	LMV358IDR / OPA1678IDR	SOIC-8	1	Rs. 18.00	Evelta / Mouser
R_BIAS1, R_BIAS2	2.2kΩ 1% — mic JFET bias pull-up to 3.3V	RC0603FR-072K2L	0603	2	Rs. 1.50	Local passives
R_FB1, R_FB2	100kΩ 1% — op-amp feedback resistor (~30dB gain)	RC0603FR-07100KL	0603	2	Rs. 1.50	Local passives
R_IN1, R_IN2	2.2kΩ 1% — op-amp inverting input resistor	RC0603FR-072K2L	0603	2	Rs. 1.50	Local passives
C_IN1, C_IN2	1μF 16V X7R — AC coupling cap	CC0603KRX7R7BB105	0603	2	Rs. 2.50	Local passives

4.2 I2S ADC — PCM1808 (Slave Mode, Pi-Synchronous Clock)

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
U_ADC	24-bit 96kHz stereo audio A/D converter, I2S output	PCM1808PWR	TSSOP-14	1	Rs. 110.00	Evelta / Mouser / DigiKey
R_FMT	10kΩ 1% — Pin 9 FMT pulldown to GND → 24-bit I2S format	RC0603FR-0710KL	0603	1	Rs. 1.50	Local passives
R_MD0, R_MD1	2× 10kΩ 1% — Pins 8/7 MD0/MD1 pulldown to GND → Slave Mode, 256fS	RC0603FR-0710KL	0603	2	Rs. 1.50	Local passives
R_BYPAS	10kΩ 1% — Pin 10 BYPAS pulldown to GND → DC high-pass filter active	RC0603FR-0710KL	0603	1	Rs. 1.50	Local passives
C_VREF	10μF 10V X5R — PCM1808 Pin 5 VREF decoupling	CL10A106KP8NNNC	0603	1	Rs. 3.50	Local passives

4.3 Clock Topology (V1.2 fix, carried through)

Pi is the I2S bus master for both the ADC and the DAC. PCM1808's SCKI (Pin 6) is driven from Pi GPCLK0 (GPIO4) configured for 12.288 MHz (256×48kHz), frequency-locking the PCM1808's delta-sigma modulator to the Pi's own audio DMA clock domain. This eliminates the two-independent-clocks risk (clicks, drift, buffer under/overflow) that exists if the ADC free-runs off its own crystal instead.

Pi Header Pin	BCM GPIO	Net Name	Connected To	Function
Pin 7	GPIO4	I2S_MCLK	PCM1808 Pin 6 (SCKI)	Synchronous system master clock from Pi GPCLK0, 12.288MHz
Pin 12	GPIO18	I2S_BCLK	PCM1808 Pin 12 (BCK)	Shared I2S bit clock, Pi master
Pin 35	GPIO19	I2S_WS	PCM1808 Pin 11 (LRCK)	Shared word-select / frame clock
Pin 38	GPIO20	I2S_DIN	PCM1808 Pin 13 (DOUT)	Digital audio input from ADC to Pi

Variant B — Cost Summary (Microphone Front End + ADC Only)

Subsystem	Included	Est. Cost (INR)
Mic Input Subsystem (Variant B)	2× electret capsules, LMV358 preamp, PCM1808 ADC, 24.576MHz Y1 (if free-run fallback kept), strapping, JSTs, passives	Rs. 172.00

5. Head Motherboard (ESP32-S3) — Corrected for Strapping-Pin Conflicts

Base architecture unchanged from the source .md (rev V1.1): camera, display, touch, tilt servo, and programming pogo-pins all on one ESP32-S3-WROOM-1-N8R8. **Three nets have been reassigned below** to remove the strapping-pin conflicts identified in Section 0 — GPIO0, GPIO3, GPIO45, and GPIO46 are ESP32-S3 boot-strap pins and must never be held in an externally-driven, ambiguous, or peripheral-controlled state at power-on/reset.

5.1 What changed vs. the source .md, and why

Signal	Old (conflicting) pin	New pin	Reason for change
TFT_DC	GPIO45 (VDD_SPI voltage strap)	GPIO40	GPIO45 selects flash voltage (3.3V vs 1.8V) at reset — a display line holding this pin in the wrong state risks unreliable boot / flash misconfiguration.
TFT_CS	GPIO46 (boot-log strap)	GPIO41	GPIO46 controls ROM boot-log printing behavior at reset; not appropriate for an externally-driven peripheral select line.
TOUCH3 (petting-A)	GPIO3 (JTAG-source strap)	GPIO42	GPIO3 selects JTAG signal source at boot. TTP223 modules default LOW when untouched, so this pin would see a defined-but-uncontrolled level every boot — moved off the strap entirely.

GPIO0 (boot mode select) and the remaining boot pins are used only for their intended purpose (R_BOOT pull-up on the pogo-pin programming header) and are not shared with any other peripheral — no change needed there.

5.2 Full BOM (unchanged from source .md unless noted)

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
U1	ESP32-S3 dual-core MCU module, 8MB flash, 8MB Octal PSRAM	ESP32-S3-WROOM-1-N8 R8	SMD Module 18×25.5mm	1	Rs. 399.00	Robu.in / Evelta / Amar Radio (Chandni)
U2	3.3V 1.0A LDO regulator (5V → 3.3V)	AMS1117-3.3 / AP2114H-3.3TRG1	SOT-223	1	Rs. 6.50	Supreme Electronics / Robu.in
C_SERVO	220µF 16V low-ESR radial — tilt servo inrush buffer	UPW1C221MPD	Radial 6.3×11mm	1	Rs. 6.00	Classic Electronics / Chandni
C1,C2,C3	10µF 16V X5R ceramic (LDO in/out, camera 3.3V)	CL10A106KP8NNNC	0603	3	Rs. 3.50	Local passives / Robu
C4-C8	100nF 50V X7R ceramic decoupling	CC0603KRX7R9BB104	0603	5	Rs. 1.50	Local passives / Robu
J_CAM	24-pin 0.5mm FPC connector (bottom-contact, ZIF)	FH12-24S-0.5SH(55)	0.5mm SMD R/A	1	Rs. 22.00	Evelta / Robu.in
CAM_MOD	OV2640 2MP camera sensor, 75mm extended flex, wide angle	OV2640-FPC-75MM	24-pin 0.5mm FPC	1	Rs. 260.00	Robu.in / Supreme Electronics
R_CAM1,R_CAM2	2.2kΩ 1% — camera SCCB/I2C pull-up (SIOD/SIOC)	RC0603FR-072K2L	0603	2	Rs. 1.50	Local passives
J_TFT	20-pin 0.5mm FPC connector (back-flip lock)	20PIN-05-FPC	0.5mm SMD R/A	1	Rs. 15.00	Robu.in / Evelta
TFT_PANEL	2.4" 320×240 ILI9341 TFT, 20-pin bare flex	ILI9341-2.4-20P-FPC	20-pin FPC panel	1	Rs. 380.00	Robu.in / Quartz / Chandni
R_BL	10Ω 1% 1/8W — backlight current limiter	RC0805FR-0710RL	0805	1	Rs. 2.00	Local passives
J_T1-J_T4	4× 3-pin JST-PH (to external TTP223 modules)	S3B-PH-K-S(LF)(SN)	2.0mm R/A	4	Rs. 7.50 ea	Evelta / Robu.in / Chandni
TP_T1-TP_T4	Circular copper touch pads (native capacitive touch alt.)	PAD-TOUCH-ROUND-6M M	PCB top copper pad	4	Rs. 0.00	PCB etched copper
TTP223_MOD	TTP223 capacitive touch breakout module	TTP223-MODULE	Mini PCB, external	4	Rs. 18.00 ea	Supreme Electronics / Chandni
R_PD1-R_PD4	10kΩ 1% — optional touch input pulldown	RC0603FR-0710KL	0603	4	Rs. 1.50 ea	Local passives

Ref Des	Component	MPN	Package	Qty	Unit Price	Source
J_SERVO	3-pin 2.54mm header — tilt servo GND/5V/PWM	1X3-PIN-HEADER-2.54	2.54mm vertical	1	Rs. 2.00	Supreme Electronics / Robu.in
SERVO_MOD	MG90S metal-gear micro servo 9g	MG90S-SERVO	External actuator	1	Rs. 125.00	Robu.in / Supreme Electronics
J_NECK	4-pin JST-PH — to Body board (5V, GND, TX, RX)	S4B-PH-K-S(LF)(SN)	2.0mm R/A	1	Rs. 9.00	Robu.in / Evelta
TP_POGO1-8	8× pogo test pads — 3V3/GND/EN/IO0/TX0/RX0/D-/D+	PAD-POGO-1.5MM	1.5mm copper pad	8	Rs. 0.00	PCB bottom etched copper
R_EN	10kΩ 1% — EN pin pull-up	RC0603FR-0710KL	0603	1	Rs. 1.50	Local passives
C_EN	1μF 16V X7R — EN power-on delay	CC0603KRX7R7BB105	0603	1	Rs. 2.50	Local passives
R_BOOT	10kΩ 1% — GPIO0 boot-mode pull-up	RC0603FR-0710KL	0603	1	Rs. 1.50	Local passives
LED1	Green LED — 3.3V power indicator	0603-LED-GREEN	0603	1	Rs. 2.00	Local passives
R_LED1	1.0kΩ 1% — LED current limiter	RC0603FR-071KL	0603	1	Rs. 1.50	Local passives

5.3 Head Board — Corrected Pin Routing Table

ESP32-S3 Pad	GPIO	Net Name	Connected To	Function
Pad 1	GND	GND	Ground plane	System common ground
Pad 2	3V3	+3V3_SYS	AMS1117-3.3 output, 10μF cap	Main 3.3V rail
Pad 3	EN	ESP_EN	Pogo TP_EN, R_EN (10k), C_EN (1μF)	Chip enable / hardware reset
Pad 11	GPIO1	TOUCH1	J_T1 Pin 3, copper pad TP_T1	Left cheek touch
Pad 12	GPIO2	TOUCH2	J_T2 Pin 3, copper pad TP_T2	Right cheek touch
—	GPIO42 (was GPIO3)	TOUCH3	J_T3 Pin 3, copper pad TP_T3	Petting-A touch — moved off JTAG-source strap pin
Pad 14	GPIO14	TOUCH4	J_T4 Pin 3, copper pad TP_T4	Petting-B touch
Pad 15	GPIO4	CAM_SIOD	J_CAM Pin 10, 2.2k pull-up	Camera SCCB/I2C data
Pad 16	GPIO5	CAM_SIOC	J_CAM Pin 8, 2.2k pull-up	Camera SCCB/I2C clock
Pad 17	GPIO6	CAM_VSYNC	J_CAM Pin 14	Camera vertical sync
Pad 18	GPIO7	CAM_HREF	J_CAM Pin 16	Camera horizontal reference
Pad 19	GPIO8	CAM_D2	J_CAM Pin 13	Camera data bit 2
Pad 20	GPIO9	CAM_D1	J_CAM Pin 15	Camera data bit 1
Pad 21	GPIO10	CAM_D3	J_CAM Pin 11	Camera data bit 3
Pad 22	GPIO11	CAM_D0	J_CAM Pin 17	Camera data bit 0
Pad 23	GPIO12	CAM_D4	J_CAM Pin 9	Camera data bit 4
Pad 24	GPIO13	CAM_PCLK	J_CAM Pin 18	Camera pixel clock
Pad 25	GPIO15	CAM_XCLK	J_CAM Pin 12	Master clock to camera (20/24MHz)
Pad 26	GPIO16	CAM_D7	J_CAM Pin 3	Camera data bit 7
Pad 27	GPIO17	CAM_D6	J_CAM Pin 5	Camera data bit 6
Pad 28	GPIO18	CAM_D5	J_CAM Pin 7	Camera data bit 5
Pad 31	GPIO35	TFT_MOSI	J_TFT Pin 5 (SDI/MOSI)	Display SPI data output
Pad 32	GPIO36	TFT_SCK	J_TFT Pin 8 (SCK)	Display SPI clock
—	GPIO40 (was GPIO45)	TFT_DC	J_TFT Pin 7 (DC)	Data/command select — moved off VDD_SPI strap pin
—	GPIO41 (was GPIO46)	TFT_CS	J_TFT Pin 9 (CS)	SPI chip select — moved off boot-log strap pin
Pad 36	GPIO47	TFT_RESET	J_TFT Pin 10 (RESET)	Display hardware reset
Pad 37	GPIO48	SERVO_PWM	J_SERVO Pin 3	LEDC PWM to MG90S tilt servo
Pad 38	GPIO43	HEAD_TX	J_NECK Pin 4, Pogo TP_TX	UART0 TX to Body Pi, 921600 baud
Pad 39	GPIO44	HEAD_RX	J_NECK Pin 3, Pogo TP_RX	UART0 RX from Body Pi, 921600 baud
Pad 29	GPIO19	USB_DM	Pogo TP_DM	Native USB D- (JTAG/flashing)
Pad 30	GPIO20	USB_DP	Pogo TP_DP	Native USB D+ (JTAG/flashing)

ESP32-S3 Pad	GPIO	Net Name	Connected To	Function
Pad 41	GPIO0	BOOT	Pogo TP_IO0, R_BOOT (10k pull-up)	Boot-mode strap (LOW = download) — unshared, no change

5.4 Camera FPC (J_CAM, 24-pin) & Display FPC (J_TFT, 20-pin) — unchanged from source .md

J_TFT — 20-pin FPC

J_CAM Pin	Signal	Connected To	Function
1,2	NC/STROBE	—	Not used
3	CAM_D7	ESP32-S3 GPIO16	Camera data bit 7
4	AGND	System GND	Analog ground
5	CAM_D6	ESP32-S3 GPIO17	Camera data bit 6
6	RESET	+3V3_SYS (via 10k)	Hardware reset, active low
7	CAM_D5	ESP32-S3 GPIO18	Camera data bit 5
8	CAM_SIOC	ESP32-S3 GPIO5	SCCB clock
9	CAM_D4	ESP32-S3 GPIO12	Camera data bit 4
10	CAM_SIOD	ESP32-S3 GPIO4	SCCB data
11	CAM_D3	ESP32-S3 GPIO10	Camera data bit 3
12	CAM_XCLK	ESP32-S3 GPIO15	Master clock to sensor
13	CAM_D2	ESP32-S3 GPIO8	Camera data bit 2
14	CAM_VSYNC	ESP32-S3 GPIO6	Vertical sync
15	CAM_D1	ESP32-S3 GPIO9	Camera data bit 1
16	CAM_HREF	ESP32-S3 GPIO7	Horizontal reference
17	CAM_D0	ESP32-S3 GPIO11	Camera data bit 0
18	CAM_PCLK	ESP32-S3 GPIO13	Pixel clock
19	DGND	System GND	Digital ground
20	AVDD	+3V3_SYS (via 10µF)	Sensor analog power
21	DOVDD	+3V3_SYS	Sensor I/O power
22	DVDD	+3V3_SYS	Internal core regulator supply
23	PWDN	GND	Power down tied active
24	GND	System GND	Ground return

J_TFT Pin	Signal	Connected To	Function
1	GND	System GND	Ground return
2	VCC	+3V3_SYS	Logic power
3,4	NC	—	Internal panel test lines
5	TFT_MOSI	ESP32-S3 GPIO35	SPI MOSI
6	NC	—	MISO not required (write-only)
7	TFT_DC	ESP32-S3 GPIO40 (corrected)	Data/command select

J_TFT Pin	Signal	Connected To	Function
8	TFT_SCK	ESP32-S3 GPIO36	SPI clock
9	TFT_CS	ESP32-S3 GPIO41 (corrected)	SPI chip select, active low
10	TFT_RESET	ESP32-S3 GPIO47	Hardware reset, active low
11	NC	—	Not used
12,13	VCC_BL	+3V3_SYS (via R_BL 10 Ω)	Backlight LED supply, fixed 100%
14-19	NC	—	Reserved / panel manufacturer lines
20	GND	System GND	Ground return

6. Consolidated Cost Summary

6.1 Body Board — Common Subsystems

Subsystem	Included	Est. Cost (INR)
Compute	Pi Zero 2W + SMD headers	Rs. 2,185.00
Power & Primary Protection	USB-C, PTC, MOSFET, 2200µF cap, TVS diode	Rs. 79.50
Servo Point-of-Load Protection	Dedicated PTC, 1000µF local cap, flyback diode, JST-XH	Rs. 36.00
Audio Output	MAX98357A, R_GAIN, ferrite beads, bypass caps, JST speaker conn.	Rs. 153.00
Peripherals & Neck Harness	Radar JST, neck JST 4-pin, shielded cable, LED	Rs. 54.50
Bare PCB (2-layer, 100×100mm batch)	—	Rs. 150.00 (approx.)
Body Common Subtotal		

6.2 Total Per-Unit Cost by Build Configuration

Line Item	Config A (INMP441)	Config B (Electret + PCM1808)
Body Common (Section 2)	Rs. 2,658.00	Rs. 2,658.00
Mic Front End	Variant A: Rs. 202.50 (INMP441)	Variant B: Rs. 172.00 (electret+PCM1808)
Head Board (Section 5)	Rs. 1,480.50	Rs. 1,480.50
TOTAL — Body + Head		

Head board cost per Section 5.2 (Rs. 1,480.50, or Rs. 1,408.50 if native capacitive touch pads are used instead of external TTP223 modules, saving Rs. 72.00). Figures are per-unit BOM cost at low-volume (5-10pc) local/import sourcing; do not include assembly, PCB batch NRE beyond the fabrication line shown, enclosure, or labor.

7. Outstanding Items Before Schematic Capture

- **Confirm C_SERVO_LOC placement order** (Section 2.3) — cap must sit electrically after F_SERVO, directly at J_SERVO pins, not before the fuse.
- **Verify GPIO40/GPIO41/GPIO42 are genuinely free** on the exact ESP32-S3-WROOM-1-N8R8 pinout revision being used (cross-check against Espressif's official datasheet pin table before finalizing footprint) — this document reassigns TFT_DC/TFT_CS/TOUCH3 off strap pins onto these, but always confirm against the current silicon rev's datasheet, not this document alone.
- **Pick ONE mic variant per production run** — Section 3 (INMP441) recommended as default; Section 4 (electret+PCM1808) only if a specific sourcing/cost reason applies. Do not lay out footprints for both unless intentionally building a populate-either-option board.
- **Touch subsystem: choose TTP223 external modules OR native ESP32-S3 capacitive pads**, not both populated simultaneously on the same net (Section 5.2 BOM lists both as population options).
- **Confirm INMP441 breakout module pinout** against the actual purchased part (6-pin breakout pinouts vary slightly by vendor) before finalizing J_MIC_L/J_MIC_R footprints in Variant A.
- **Run a servo brownout bench test** on the new point-of-load protection branch before committing to the full production panel — this is a previously-occurred failure mode (documented ENABLE_SERVOS=0 workaround in the current build) and deserves empirical verification, not just schematic review.
- **Confirm CM4 antenna/U.FL requirement is moot** for this run (Pi Zero 2W only, per this session's direction) — flagged here only so it isn't silently forgotten if a CM4 variant is revisited later.